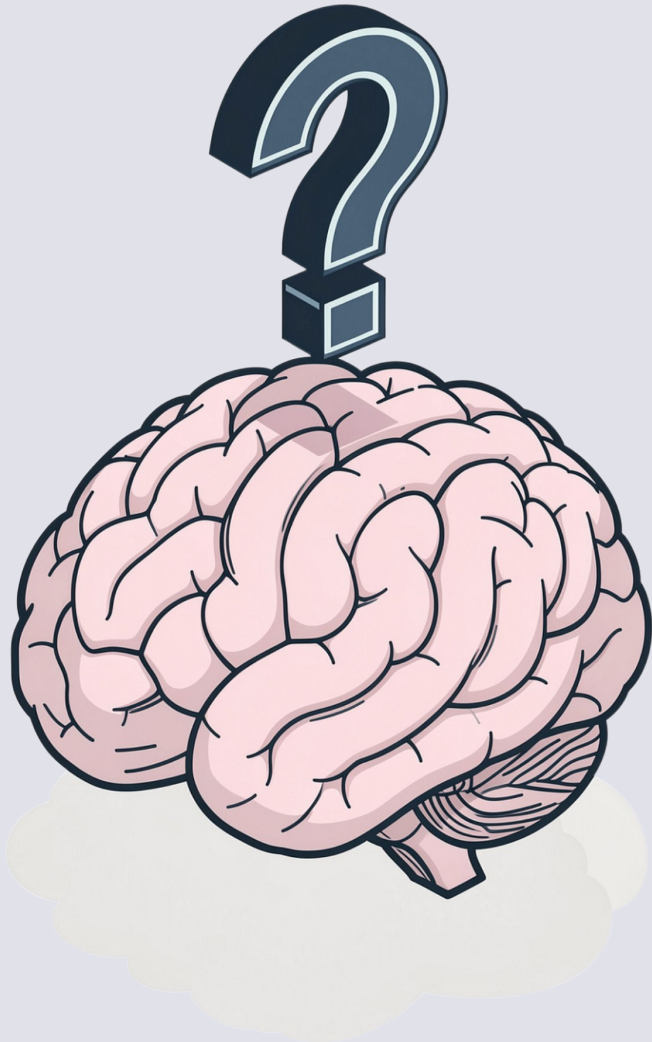


Memory Limitations of LLMs & Agent Memory

Unit 3 — Part 1: Understanding how LLMs process information and how agents retain knowledge across conversations.





Does an LLM Actually Have Memory?

No Permanent Memory

LLMs don't have unlimited, permanent memory of everything you've told them.

Context-Driven Responses

They generate responses based on information available in their current context.

Recall Requires Availability

Relevant conversation must still be available in context to recall it accurately.

The Context Window



Definition

Maximum amount of input/context the model can process in a single interaction.

Includes

- System instructions
- User messages & previous conversation
- Retrieved documents & tool results

Analogy

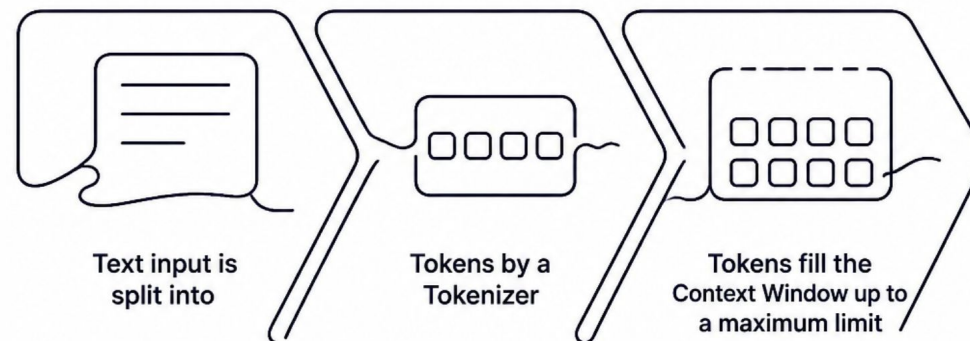
A desk with limited space — eventually you can't fit everything.

Tokens: What's Actually Limited

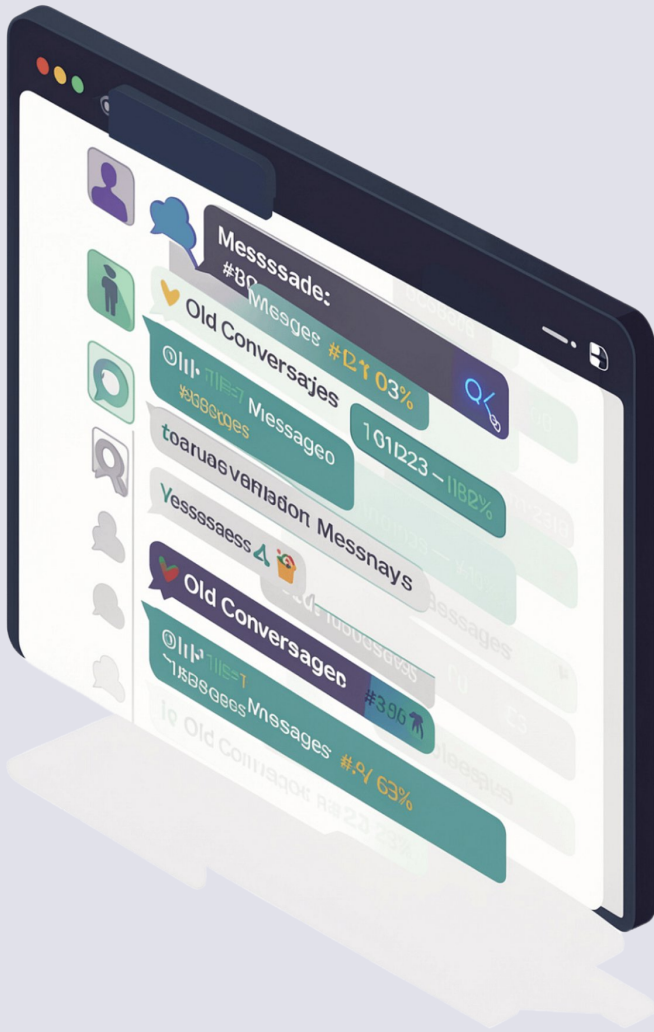
How LLMs Measure Context

LLMs measure context in **tokens**, not sentences. Text is broken into tokens by the model's tokenizer — words, subwords, and punctuation all count.

Context length = maximum tokenized information the model can process in one interaction.



The Long Conversation Problem



Truncation

In very long conversations, older information may be truncated, summarized, or selectively retrieved.

Unlimited History Is a Myth

The model cannot treat unlimited history as available context — there is always a ceiling.

Result

Information from earlier messages becomes unavailable or diluted over time.

What "Forgetting" Actually Means

Not This

LLMs have human-like memory and forget things the way people do.

Actually This

Previously provided information may no longer be **available in the model's active context** due to context limitations and retrieval strategies.

It's an architectural constraint — not a cognitive one.

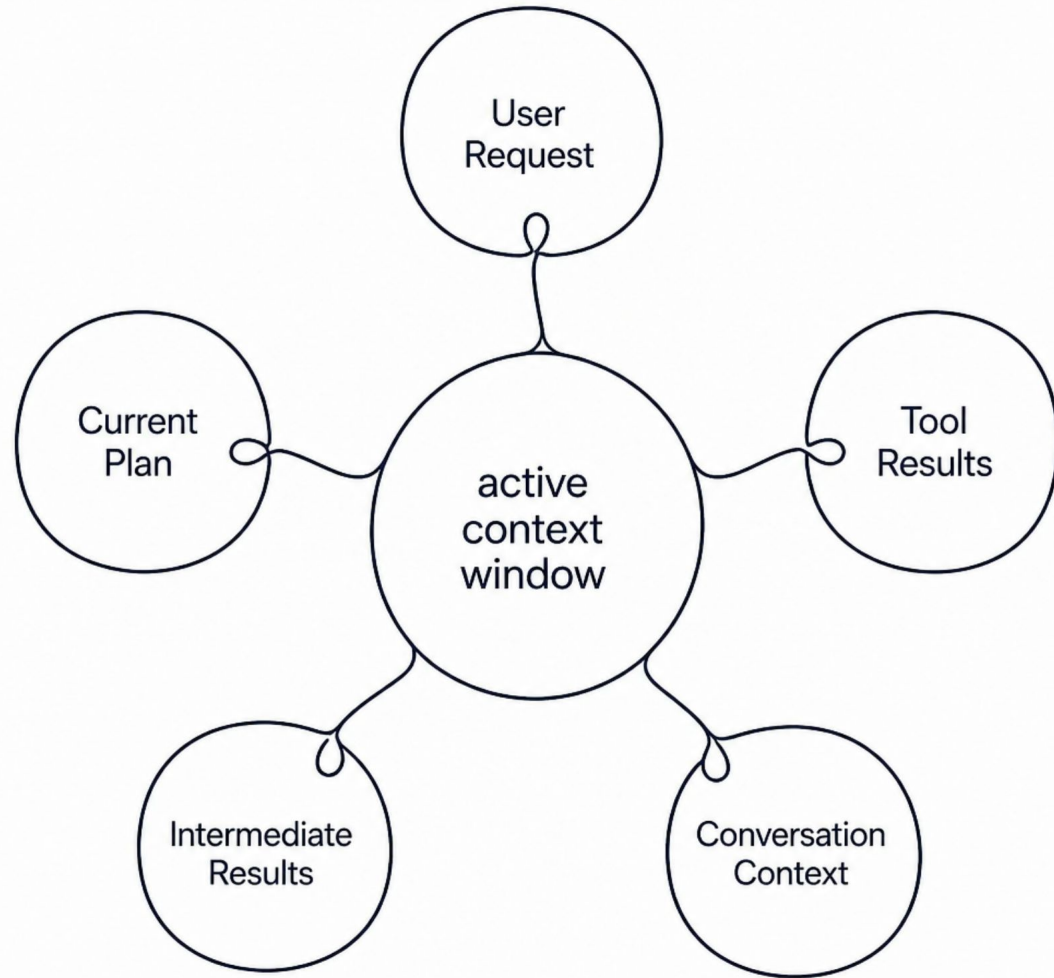
The Agentic AI Problem

If an agent works for **hours, days, or across multiple conversations**, you cannot simply keep putting everything into the context window.

- 1 Agent memory architecture becomes essential
- 2 Short-term and long-term memory strategies matter



Short-Term Memory



Definition

Information relevant to the **current conversation or current task**.

Includes

- User request
- Current plan
- Tool results
- Intermediate results
- Conversation context

📌 **Key Question:** "What do I need to know right now?"



Long-Term Memory

Definition

Information retained for **future interactions** — persisted beyond the current session.

Example

User preferences like *"window seats," "economy class," "morning flights"* — stored and recalled in later conversations.

Key Question

"What information should I retain for future interactions?"

Don't Confuse State and Memory

State

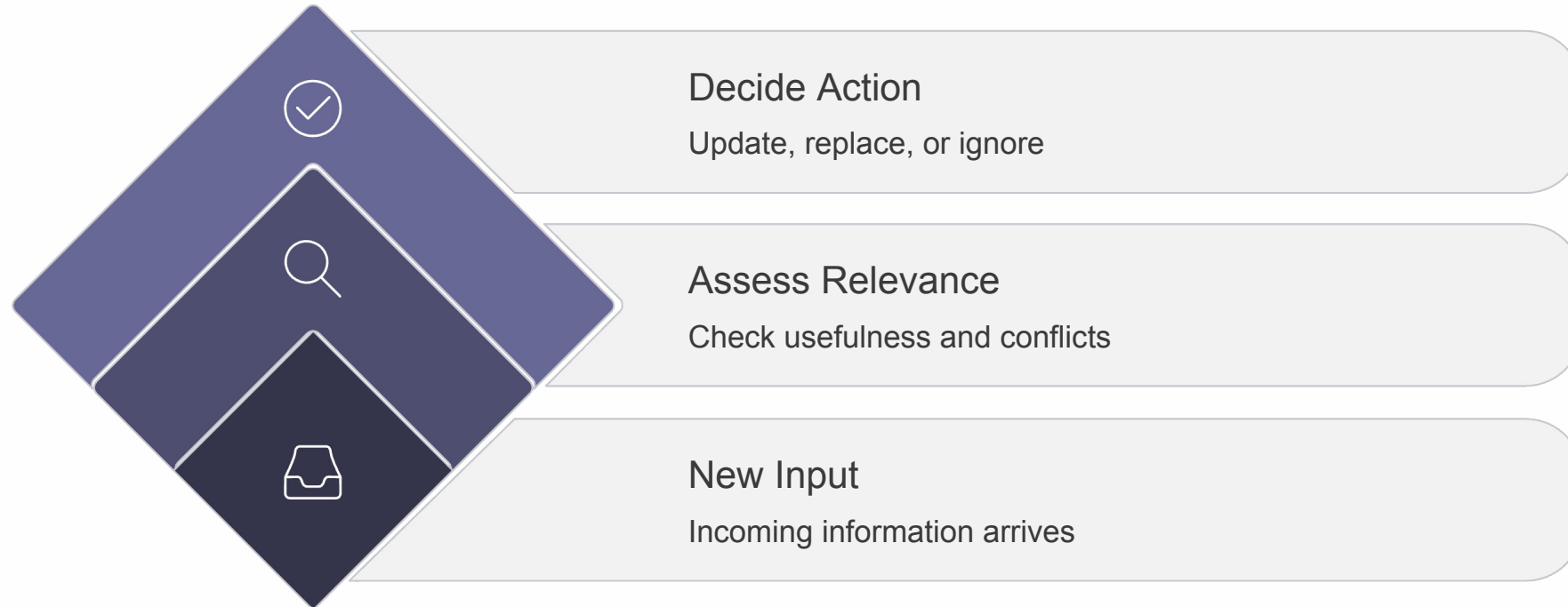
Current task information — destination, date, number of passengers. Tells the agent **where it is** right now.

Memory

Persistent information — preferences, past facts. Tells the agent **what it remembers** across time.

📌 **Key Distinction:** State tells the agent where it is; memory tells the agent what it remembers.

Memory Requires Decision-Making



Is it useful?

Should this new information be stored at all?

Does it conflict?

Does it contradict what the agent already knows?

What action?

Update, replace, or ignore — based on relevance and accuracy.